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# Our World – Our Goals 2.0

## *Updated Recommendations on Strengthening Citizen Science for the SDGs*

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# Table of contents

Taking Stock: Progress Made_____ 3

References_____ 9

12 Updated Recommendations on Strengthening Citizen Science for the SDGs_____ 14

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# Taking Stock: Progress Made

In 2015, the Sustainable Development Goals (SDGs) were adopted by the UN as part of the 2030 Agenda for Sustainable Development. They comprise 17 goals and 169 targets, which are monitored through 231 unique indicators. While methodology development and global-level reporting is in the hands of UN agencies or other international organizations (so-called custodian agencies), the bulk of SDG monitoring is undertaken on the national level, by National Statistical Offices (NSOs) and other relevant public authorities. As the latest SDG report concedes, **progress towards achieving the 17 goals so far has been "deeply inadequate,"** both on the level of monitoring and on the level of actual implementation (UN Department of Economic and Social Affairs, 2025)**. Citizen science can contribute to overcoming these shortcomings** not only by generating data to fill monitoring gaps, but also by fostering concrete action on the ground. Citizen science initiatives are helping to co-create solutions, raise awareness, and drive behavioral change in local communities. Involving individuals and communities in both official data ecosystems and local action ensures approaches are inclusive and responsive to societal needs and priorities. In this way, citizen science supports the SDG commitment to 'leave no one behind' and rises to the call for "responsive, inclusive, participatory and representative decision-making at all levels" (SDG 16.7). **Data produced by citizen science initiatives (a key source of citizen-generated data)[1] is moving beyond the stage of a 'pilot curiosity' and showing signs of recognition within the global data-value chain–including the SDG reporting ecosystem**. The Global Sustainable Development Report 2023 states that "novel partnerships and data technologies – for example, philanthropic funding, citizen science, artificial intelligence - must be used in the second half of Agenda 2030" to address data gaps and inform action (United Nations, 2023).

**Box 1: Progress Highlight**

The Ghana Statistical Service (GSS) worked with stakeholders to create a three-step workflow for the validation of citizen science data. The success of this case study was aided by an update of the Statistical Service Act in 2019 - creating an enabling environment for more inclusive data ecosystems. Political support and the commitment and willingness of citizen scientists to work with government representatives and international partners was also crucial (Fraisl et al., 2023b).

In New Zealand, a citizen science initiative has been implemented to collect beach litter data to support monitoring for indicator 14.1.1b (Ihimaera-Smiler, 2020). Citizen observations are also part of the EU SDG indicator framework monitoring for SDG 15 (Life on Land) through the European Butterfly Monitoring Scheme and the Pan-European Common Bird Monitoring Scheme (European Commission Joint Research

[1] Berti Suman et al. (2020) define citizen-generated data as "data produced through citizen science, sensing and other forms of civic monitoring that share the common denominator that the data collection process is primarily carried out by volunteer citizens actively joining the initiative." Fritz et al. (2019) frame it as "data produced by citizens and their organizations in monitoring issues that affect them in order to realize change. The concept of citizen-generated data overlaps with many other terms including citizen science[…]".



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Centre & Knowledge Centre for Biodiversity, European Environment Agency, 2025). All monitoring data on birds compiled by the International Union for Conservation of Nature and Natural Resources under SDG indicator 15.5.1 ('Red List Index') is provided by BirdLife International with its large network of trained volunteers (Fraisl et al., 2020; Fritz et al., 2019).

The global methodology for SDG indicator 14.1.1b (*Plastic Debris Density)* explicitly recommends citizen science as a suitable data source (UNEP, 2021). National statistical offices in Ghana, Colombia, the Philippines, Mexico, the UK, Kenya and Italy have explored the use of non-traditional data sources (Proden & Imaralieva, 2021). The Philippine Statistics Authority, for example, uses household census data collected by volunteers to complement their statistics on 32 SDG indicators (Fritz et al. 2019).
In 2022, Ghana officially integrated citizen science data on marine plastic litter into its monitoring and reporting of the abovementioned SDG indicator 14.1.1b for the years 2016–2020 (Fraisl et al., 2023b). In addition, data is being gathered data via a Public Services Satisfaction App (PSS) to contribute towards reporting of SDG Indicator 16.6.2[2]. The UNDP Accelerator Labs, an initiative launched in 2019 to tackle wicked sustainable development challenges and accelerate progress towards the SDGs, have implemented citizen science activities on topics like air pollution and water resource management in Argentina, Bolivia, Guatemala, Panama, and India (UNDP Accelerator Labs, 2023; UNDP Accelerator Labs, 2024; UNDP Accelerator Labs, 2025a; UNDP Accelerator Labs, 2025b). They have also recently published a comprehensive report on the added value of citizen science and other forms of collective intelligence in the context of climate change mitigation and adaptation (Berditchevskaia, A. et al., 2024).

**Box 2: Untapped potential in water quality monitoring**
Isabel Bishop and colleagues (2020) have assessed how citizen science can support national reporting on indicator 6.3.2, which tracks the quality of water bodies. They found that CS can complement official monitoring, particularly where resources and infrastructure are limited. Projects like FreshWater Watch (FWW) use low-cost tools and validated methods to generate reliable data. CS also supports SDG goal 6b by engaging communities and raising awareness. However, challenges remain: lack of sustained funding, limited stakeholder engagement, and uneven data coverage can limit integration into official reporting.

**Two governance milestones in 2023 anchor this global momentum.** In November 2023, the UN DESA Statistics Division released the *Copenhagen Framework on Citizen Data*, providing national statistics offices (NSOs) a shared vocabulary for classifying, validating, and scaling citizen inputs. Earlier, at the UN World Data Forum in April 2023, the *Collaborative on Citizen Data* was formally launched, establishing a standing forum where governments, NGOs, and tech providers co-design pilots and share lessons learned.

**Yet, in spite of all this progress, a lot of untapped potential remains.** Around 30% of SDG indicators still suffer from poor data availability (UN

[2] The app and pilot programme was jointly developed by the United Nations Development Programme (UNDP) and the Ghana Statistical Service (GSS). See https://www.undp.org/policy-centre/governance/news/pioneering-change-ghanas-bold-leap-citizen-science-public-service-satisfaction.


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DESA, 2025). Even for the roughly 70% with good coverage, monitoring data is often outdated or fails to meet granularity standards formulated by the United Nations Statistics Division (OECD, 2025b). Citizen science has to be part of the solution: A systematic inventory of CS contributions to the SDGs undertaken in 2020 found that, apart from those it was already providing data for, CS projects could contribute to an additional 76 indicators (Fraisl et al., 2020). A more recent study found that citizen science has the potential to contribute to 85% of all indicators under SDG 3 (Good Health and Well-being) (Fraisl et al. 2023a). Other studies have proposed concrete strategies for leveraging citizen science initiatives or existing citizen-generated datasets to complement the monitoring of specific SDG indicators (cf. example boxes in this document).

**Box 3: CS for soil-related SDG targets**

Head et al (2020), analysed the role that CS can play in achieving SDG targets related to soil health, which has long been underrepresented in the wider discussion around sustainability. As a result, standardised metrics for soil monitoring are lacking. Incorporating complimentary approaches is therefore crucial to meet soil-related SDG targets by 2030. Beyond closing data gaps, citizen science initiatives can also benefit key target groups, such as farmers, if they combine data collection with advisory services and practical recommendations. While a number of CS methods for soil monitoring have been successfully validated, some barriers remain: concerns about data quality and reliability, time and resource-intensive sampling methods, and lack of project sustainability.

**Apart from contributing to the monitoring of specific SDG indicators, citizen-generated data is becoming statistically visible evidence that can trigger action towards advancing the SDGs on the level of goals and targets.** Under the Swachh Bharat Mission (SBM)-Urban 2.0[3] for example, Indian cities are ranked for cleanliness as part of a national effort to make all cities "Garbage Free" and contribute to the UN SDG 2030 goals. A significant feature of this initiative is active citizen participation through digital platforms. Apps empower citizens to collect sanitation-related data and report grievances, thus supporting improved sanitation and urban governance. In addition to collecting data, citizen science initiatives can help implement the SDGs on the goal and/or target level by raising awareness, fostering behavioral change, building connections between different stakeholder groups, and co-designing local solutions (Sauermann et al., 2020; Queiruga-Dios et al. 2020; Ajates et al. 2020).
Of course, the extent to which citizen science can realize its potential for the SDGs ultimately depends on the maturity of the field as a whole. Progress in mainstreaming citizen science across disciplines, institutions, and policy arenas directly shapes its capacity to deliver on global sustainability goals.

One important indicator of this maturity is the recognition of citizen science in formal policy contexts. The growing interest in citizen science within established policy illustrates its increasing perception as a valuable resource for evidence-based decision-making. For instance, a recent policy paper produced by

[3] https://sbmurban.org/digital_Inovation

the OECD Global Science Forum (GSF) stresses that citizen science can boost data coverage, accelerate scientific discovery, help tackle societal challenges, and increase the legitimacy and uptake of evidence-based policies (OECD, 2025a). The European Network of Heads of Environmental Protection Agencies (EPA Network) hosts an informal Interest Group on Citizen Science, which has published a set of recommendations on the use of citizen science for environmental monitoring (Interest Group on Citizen Science within the EPA Network, 2022). Leveraging the full potential of citizen science is also part of strategic objective 4 ("making full use of the potential of data, technology and digitalisation"), of the EEA-Eionet strategy 2021-2030 (European Environment Agency, 2021).

At the same time, the credibility and usability of citizen-generated data remains a central concern, especially for researchers and policy makers. Like other types of monitoring data, citizen-generated data faces challenges around data management, including standardization, harmonization, aggregation, and accessibility (European Commission Joint Research Centre, 2025; Lumbierres et al., 2025; Oturai et al., 2023). Concerns about their reliability and accuracy also continue to present a significant obstacle to the uptake of citizen-generated data by policymakers and NSOs (OECD 2025a; Proden and Imaralieva, 2021; de Sherbinin et al., 2021). Overcoming this skepticism will require a combination of advocacy, capacity-building among CS practitioners (e.g. on applying quality assurance strategies proposed by Fraisl et al., 2022 and Johnston et al., 2023), and close coordination between data producers and data users. After all, data quality is not an absolute value but a measure of 'fitness for use', i.e. it depends on the intended application (Gumiero et al., 2025; Bowser et al., 2020). One strategy for accelerating the acceptance of citizen-generated data in policy contexts could consist in updating the Aarhus Convention, a United Nations treaty granting the public rights to access environmental information, participate in environmental decision-making, and seek justice in environmental matters adopted in 1998, to include a duty for authorities to consider environmental information produced by citizens if it meets pre-defined quality standards–a possibility that was briefly explored during the 7th Meeting of the Parties (MoPs) of the Aarhus Convention in 2021 (Berti Suman et al., 2023).

Beyond questions of data quality, another challenge is the fragmentation of initiatives and resources. The steadily growing interest in CS during the past decade has led to a proliferation of projects, initiated and managed by a range of organizations and often operating with short-term funding and limited reach. This was also emphasized as a key problem during the ECS Cluster Event 2025 (Torres & Schuerz, 2025). Considerable efforts have already been made to address the resulting fragmentation of citizen science initiatives and their outputs. The Mosquito Alert project has developed a global platform to align citizen science mosquito monitoring efforts around the world. Building on lighthouse examples such as the Plastic Pirates, several Horizon Europe projects (e.g. ScienceUs, OTTERS, and CROPS) are investigating how citizen science activities can be successfully scaled up. With its web platform https://citizenscience.eu/, ECS is serving as a networking and knowledge-exchange hub for the community, while RIECS-concept will develop the concept of an integrated pan-European research infrastructure for citizen science. The Citizen Science Global Partnership (CSGP), founded in 2022, connects existing citizen science networks with actors representing policy, business, and civil society. Continued policy support and funding will be needed to further improve strategic coordination and foster

sustained collaborations, thereby maximising the impact of citizen science within and beyond the area of sustainable development.

The emergence of numerous small-scale, short-lived citizen science projects has also resulted in a fragmentation of data, as different projects often deposit their datasets in disparate repositories, hampering data integration and comprehensive analysis (Hansen et al., 2021; Bowser et al., 2020). However, considerable inroads have been made in this respect as well, for instance through the integration of citizen-generated data into data infrastructures like GBIF, EMODnet, and EASIN. The Horizon Europe project MoRe4nature is actively advancing the uptake of citizen-generated data and citizen actions in environmental compliance assurance by fostering collaborations between citizen science initiatives and public authorities, developing data validation tools, and promoting the integration of citizen-generated data in the European Open Science Cloud, the Green Deal Data Space. To build on this momentum and further increase the amount of citizen-generated data shared and aggregated through such platforms, sustained support will be needed to raise awareness among citizen science practitioners, provide training, and facilitate dialogue between relevant stakeholders.

In addition to tackling fragmentation, embedding citizen science more firmly in the higher education and research system is a key priority. Numerous EU-funded projects and initiatives have promoted the mainstreaming of citizen science across the science system through advocacy and capacity-building. Examples include TIME4CS, LibOCS, CIVIS, and of course ECS itself through its ECS Academy and collaboration with early career researchers within the Marie Curie Alumni Association. The ProBleu project engages schools in citizen science activities to boost ocean and water literacy. Still, a sustained effort will be necessary to really embed citizen science into school and university curricula.

Another factor limiting the capacity and impact of citizen science is a lack of diversity. Engagement varies significantly among countries, regions, and demographic groups, with individuals and communities who are most affected by and have valuable knowledge on issues such as environmental pollution and public health still frequently left out (Pateman and West, 2023). Building on the learnings of EU-funded projects like WeCount, IMPETUS, and ACTION, citizen science can enhance the relevance, and societal impact of research and innovation to better address pressing societal goals - but only if adequate and sustained funding is available to support inclusive engagement.

Lastly, disciplinary divides influence the uptake and visibility of contributions from citizen science. While natural sciences tend to lead in the development and application of citizen science methodologies, often with more clearly defined indicators, similar practices exist within the social sciences and humanities. Moreover, citizen engagement in research is carried out under a variety of different labels (such as crowdsourcing, participatory action research (PAR), community-based monitoring, or co-creation) depending on disciplinary traditions or institutional contexts. While this diversity reflects the richness of approaches to engaging citizens in knowledge production, it also risks fragmenting recognition and policy impact. To maximize synergies and visibility, it is therefore helpful to consider these approaches together.

**In summary, citizen science has made important strides in visibility, participation, and thematic alignment with the SDGs. Nevertheless, significant work remains to ensure that citizen-generated data**

**is effectively integrated into official statistics and monitoring frameworks, used to inform decision-making processes, and translated into action to advance sustainable development.** Moving forward, there still remains a clear need for more coordinated approaches that emphasize methodological rigor, data interoperability, capacity building, and policy relevance. Increased investment in infrastructure, institutional frameworks, and cross-sector collaboration will be essential to unlocking the full potential of citizen science in advancing the 2030 Agenda for Sustainable Development.


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# 12 Updated Recommendations on Strengthening Citizen Science for the SDGs

As the previous section of this document has demonstrated, citizen science has proven potential to complement official statistics and accelerate progress on the Sustainable Development Goals. To fully realize this potential, policymakers, national statistical offices, research institutions, and civil society need to work together on four key fronts: data credibility, data uptake, capacity and collaboration, and policy alignment and long-term support. The policy recommendations presented here were iteratively developed by synthesizing insights from a co-design process (including workshops and asynchronous collaboration via Padlet and an online living document), discussions at a high-level policy event we hosted in January 2025, and expert feedback on the draft document.

**Policy Recommendations**

1. Increase Trust in and Legitimacy of Citizen-Generated Data

- **Strengthen awareness and advocacy**: Launch coordinated campaigns that showcase successful uses of CS in SDG monitoring and provide shared resources, messaging, and visibility, while enabling individual countries to adapt materials to their specific contexts. Target audiences should include policymakers, researchers or experts informing policymaking, and NSOs, as well as citizen science practitioners and researchers.
- **Establish trusted and transparent data sources:** Support and fund transparent, quality-assured repositories for citizen-generated data which are aligned with FAIR principles and safeguard data integrity through appropriate security measures, promoting the re-use of existing good practices and infrastructures. Encourage CS projects to, where appropriate, adhere to FAIR and CARE standards and publicly document their methodologies, limitations, and uncertainties, supported by dedicated guidelines and training.
- **Develop quality assurance and data validation frameworks**: Promote knowledge-sharing to compile and consolidate existing quality assurance protocols and data validation frameworks. Where necessary, close gaps through co-design, involving statisticians, CS practitioners, and third-party experts familiar with both strategic-level data quality demands and the working environment of citizen scientists on the ground to ensure that the results reflect the needs and priorities of both data producers and data users.


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2. Enhance Usability and Uptake of Citizen-Generated Data

- **Create institutional workflows and standardized methodologies:** Facilitate the integration of citizen-generated data into official SDG reporting and other policy-relevant statistics by establishing clearly defined institutional pathways and workflows, and by disseminating standardized methodologies and toolkits for the CS-based monitoring of high-priority SDG indicators.
- **Align CS research activities with policymaking needs:** Provide support and guidance for CS initiatives to design and implement their research activities with policymaking in mind, for instance by disseminating overviews of policy-relevant data gaps and information about relevant knowledge brokers (e.g. advisory bodies), data repositories, metadata standards etc.
- **Promote user-friendly visualization and translation tools**: To improve accessibility for non-technical audiences (including policymakers), fund the creation of easy-to-use tools, such as dashboards that visualize data and allow policymakers to download filtered subsets, as well as the development of plain-language summaries, ideally produced by CS practitioners who can provide accurate and context-sensitive translations, complemented where useful by automated tools.

3. Strengthen Capacity and Foster Collaboration

- **Invest in training and infrastructure**: Develop training and mentoring schemes on the SDGs, data management and policy engagement for CS practitioners across all academic fields and disciplines, including the social sciences and humanities (for instance in the form of MOOCs co-developed with the ECS Academy), support NGOs and local-level institutions in delivering community-based capacity development, and invest in technical infrastructure and expertise to ensure data quality and usability.
- **Foster sustained dialogue and networks:** Strengthen existing CS networks (such as the Citizen Science Global Partnership, ECS collaboration group and network of CS ambassadors) to facilitate collaboration, scaling, and peer-learning, include CS representatives in international and national SDG monitoring forums, and develop platforms (e.g. annual stakeholder cafés) for regular dialogue between NSOs, policymakers and CS projects to help align their priorities and close current gaps in coordination.
- **Engage youth and early career researchers**: Advance the integration of CS and the SDGs into school and university curricula to raise awareness and increase the visibility and legitimacy of CS for sustainability, positioning youth as drivers of transformative change, strengthen existing networks at the interface of sustainability research and education, and support the reform of research assessment to incentivize researcher engagement in citizen science initiatives.


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4. Mainstream Citizen Science through Policy, Funding, and Broad Engagement

- **Update legal and policy frameworks:** Advance legal reforms that mandate statistical agencies to consider CS as a data source for official reporting if quality criteria are met, and promote the embedding of CS in national-level open science, open data, sustainability, and digital governance strategies.
- **Provide dedicated and long-term funding:** Foster the creation of dedicated funding mechanisms for CS projects contributing to the SDGs and sustainability more generally, e.g. in the context of national and regional green transition funds, including accessible funding modules for local, small-scale, and grassroots CS projects; encourage member states to support cross-border consortia by offering matched funding and shared infrastructure grants to cover the costs of facilities or platforms used collaboratively across borders; develop longer-term funding mechanisms (including funding for umbrella organizations like ECSA) to help improve coordination, consolidate partnerships, facilitate capacity-building, and increase impact.
- **Widen participation in research and agenda-setting:** Ensure local communities and underrepresented or hard-to-reach groups affected by the issues under investigation and/or holding unique knowledge about them are meaningfully involved not only in SDG-related citizen science, but also in the drafting of new sustainability frameworks, respecting Indigenous knowledge under CARE (Collective Benefit, Authority, Responsibility, Ethics) principles.


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